

Jason Nguyen

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[LinkedIn Profile](#) | [Personal Website Portfolio](#) | [GitHub Profile](#)

CS graduate with research and internship experience spanning ML pipelines, distributed data systems, embedded systems, and full-stack development.

SKILLS

ML/AI: PyTorch, TensorFlow, PySpark, OpenCV, NumPy, Pandas

Web/Full-Stack: React, Flask, Django, HTML, CSS, Node.js, Svelte

Languages: Python, Rust, Java, JavaScript, TypeScript, C/C++, C#, Go, Scala, GDScript

Tools & Platforms: Git, GitHub Actions, Bash, MySQL, SQLite, PostgreSQL, MongoDB, Azure, Google Cloud, Arduino, Figma, JIRA, Docker, Kubernetes

EDUCATION

SUNY University at Buffalo

Bachelor of Science, Computer Science, 2026

Cicero-North Syracuse High School

Graduated with Honors, 2021

EXPERIENCE

CACI International Inc

June 2026 - Present

Full Stack Software Developer

FedEx

January 2026 - May 2026

Customs Trade Specialist

- Built Excel automation workflows with Power BI, reducing manual customs processing time by 30%, improving throughput for time-sensitive international shipments.
- Leveraged SQL, Excel reporting, and cross-platform brokerage systems to increase daily volume by 25%.

ARISE Adaptive Design

May 2024 - August 2024

Adaptive Design Intern

- Engineered and validated object-oriented firmware for 2 embedded systems (Arduino/C++) in a Linux environment, covering 23 unit test cases across sensor input, signal timing, and hardware-software integration.
- Reduced latency by 15% through iterative firmware optimization, using serial debug logs to isolate and resolve bottlenecks.
- Authored hardware configuration and debugging documentation that standardized setup across 3 device variants, cutting team onboarding and reconfiguration time.

Randstad

January 2024 - April 2024

Data Analyst Intern

- Cleaned and validated 9,000+ records using Excel (pivot tables, VLOOKUP, conditional logic) and SQL, resolving 4 categories of recurring data entry errors.
- Built 2 reporting templates that automated manual aggregation, reducing report turnaround by 1 hour per cycle.
- Conducted root cause analysis on data workflows and produced reports to reduce errors in weekly deliverables.

Booz Allen Hamilton

May 2023 - August 2023

Software Engineer Intern

- Built REST API endpoints and SQL queries, using React and Flask, for internal dashboard and reporting systems replacing manual processes to reduce processing time by 15% for a team of 15 analysts.
- Delivered 5 sprint-based features across a 3 month engagement using Git, JIRA, Docker, and CI/CD pipelines.

DataAnnotation

May 2022 - August 2022

Freelance Software Engineer

- Reviewed 500+ AI-generated outputs using structured evaluation rubrics, documenting failure modes across edge cases to inform model fine-tuning.
- Designed reusable prompt templates and edge case test suites across 3+ task types, standardizing evaluation criteria for LLM code generation benchmarks.

RESEARCH

Large-Scale Spotify Track Popularity Analysis with PySpark & ML | [Link to Report](#)

- Designed and executed a large-scale data analysis and machine learning pipeline on 97,624 Spotify tracks using PySpark on HDFS. Built a Random Forest Regressor to predict song popularity from audio features; improved R^2 by 41% (0.17 to 0.24) through hyperparameter tuning via 3-fold cross-validation. Performed genre popularity distribution analysis across 114 genres and audio feature significance testing (Welch's t-test, Cohen's d) across six popularity tiers, identifying acoustiness and danceability as the strongest predictors of viral performance.

Handwriting to Text: CRNN + LLM Study Guide Pipeline | [Link to Report](#)

- Built an end-to-end pipeline that converts handwritten student notes into structured study guides. Trained a Convolutional Recurrent Neural Network (CRNN) with CTC loss on 400k+ word images, learning character-level feature embeddings for sequence recognition, from the IAM Handwriting Database and Kaggle, achieving 12.47% Character Error Rate. Integrated Llama 3.3 70B via the Groq API for downstream summarization, reaching ROUGE-1 F1 of 0.423. Evaluated against Tesseract and BLIP-2 baselines.

PROJECTS

Personal Portfolio Website | *React, JavaScript, CSS, HTML, Git, Vercel* | <https://website-plum-six-56.vercel.app/>

- Built and deployed a personal portfolio website using React and Vercel, implementing a responsive single-page application with custom CSS styling and component-based architecture, serving as a centralized showcase of projects and professional background.

BDO Loot Tracker | *Python, Svelte, CSS, SQLite, Git* | <https://github.com/janhnguyen/BDO-Loot-Tracker>

- Built a desktop app in Python and Svelte, capable of handling 10,000+ tracked items with local SQLite persistence, and set up automated build and release pipelines via GitHub Actions, resulting in 40+ downloads.

Lost Ark Logs | *Rust, Svelte, TypeScript, SQLite, Git* | <https://github.com/snoww/loa-logs>

- Contributed UI/UX components and SQLite query optimizations to an open-source analytics tool serving 5,000+ concurrent users, improving load times by 50%, and owned features end-to-end across code reviews, issue tracking, and iterative releases in a multi-developer Git workflow.

Tic-Tac-Toe Bot | *Python, PyTorch* | <https://github.com/janhnguyen/rl446tictactoe>

- Developed a reinforcement learning framework in Python using PyTorch, implementing DQN, PPO, SAC, and Rainbow agents with a custom Tic-Tac-Toe environment, reward functions, and training pipeline to evaluate agent performance and learning efficiency.

OCR Party Inspection | *Python, PyTesseract, OpenCV, Numpy* | <https://github.com/janhnguyen/partyInspection>

- Built a Python application using OpenCV and PyTesseract to capture and process game client screenshots, extract structured data via OCR and image recognition, and transmit results to an external web service through automated data pipelines.

Social Media Site | *React, CSS, JavaScript, HTML, Git, Docker, Kubernetes* | <https://github.com/janhnguyen/djibouti>

- Developed a full-stack social media web application using React and a custom backend API, implementing responsive UI components, RESTful client-server communication, and containerized deployment with Docker and Kubernetes to ensure scalability, portability, and maintainable architecture.

LEADERSHIP & COMMUNITY INVOLVEMENT

Lost Ark Nexus Contributor | <https://www.lostark.nexus/> | <https://shorturl.at/18Kle>

- Authored and maintained a comprehensive technical guide analyzing system mechanics, optimization strategies, and performance tradeoffs for a large online gaming community.
- Interpreted gameplay metrics and iterative updates to refine recommendations. Actively supported users through Q&A and technical review.

UB Open Source Club

- Collaborated as an open source club member to develop indie games, contributing to coding, design, and testing, which enhanced project quality and innovation. Participated in brainstorming sessions and provided feedback.